

prerequisites · Arrays

Arrays

L-1

L-2

L-3

L-4

# 2D Arrays & 2D Vectors

Leetcode  
ques lists

Advance  
Arrays

Prefix, Sliding  
Kadane, Voting

Summary Notes

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## Ques: Merge 2 Sorted Arrays

$TC = O(m+n)$

a

|    |    |    |    |    |     |
|----|----|----|----|----|-----|
| 10 | 20 | 40 | 70 | 90 | 100 |
|----|----|----|----|----|-----|

b

|    |    |    |    |
|----|----|----|----|
| 30 | 50 | 60 | 80 |
|----|----|----|----|

i

j

c

|    |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|-----|
| 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
|----|----|----|----|----|----|----|----|----|-----|

k

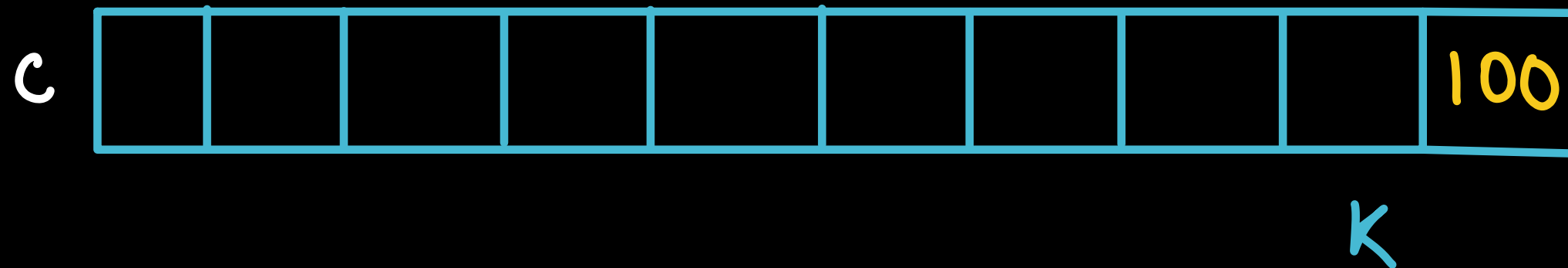
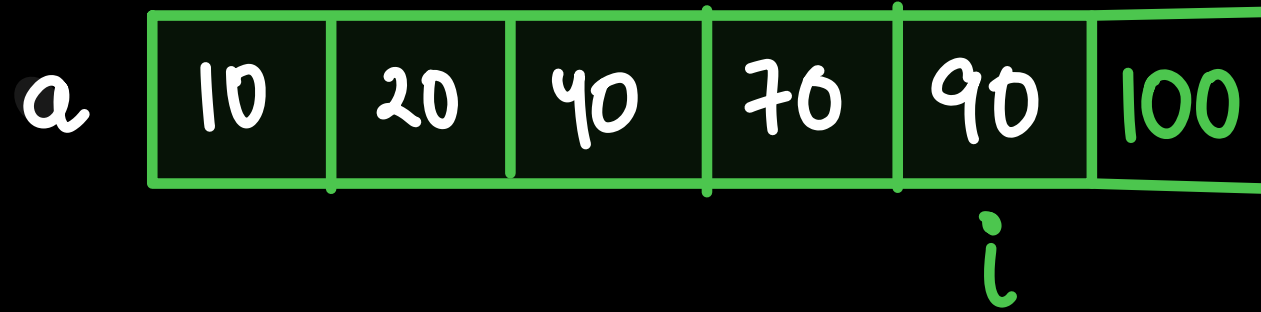
```
if(a[i] < b[j]){  
    c[k] = a[i];  
    i++; k++;  
}
```

**HW** **M-2**  
**Ques:** Merge 2 Sorted Arrays



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$$TC = O(m+n)$$



# 88. Merge Sorted Array

'Homework'



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# Contents

- ✓ 1. Representation, Creation & Indexing
2. Traversal
3. Questions
4. ArrayList of Arraylist
5. Advance Questions

# Representation of 2D Array / Matrix

|   | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 9 | 4 | 8 | 4 |
| 1 | 6 | 1 | 1 | 5 |
| 2 | 2 | 2 | 3 | 6 |

arr 3x4

$arr[row][col]$

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 4 | 3 | 6 | 9 | 8 | 1 |

$arr[idx]$

# Representation of 2D Array / Matrix

Array of Arrays

|   | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 9 | 4 | 8 | 4 |
| 1 | 6 | 1 | 1 | 5 |
| 2 | 2 | 2 | 3 | 6 |



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$$\text{arr} = \left\{ \overset{0}{\{9, 4, 8, 4\}}, \overset{1}{\{6, 1, 1, 5\}}, \overset{2}{\{2, 2, 3, 6\}} \right\}$$

$\text{arr}[1][2]$

# Input & Output in 2D Arrays

TC =  $O(mn)$

How to print elements column wise  
Transpose Print karo

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |

→

|   |   |
|---|---|
| 1 | 5 |
| 2 | 6 |
| 3 | 7 |
| 4 | 8 |



**Ques:** Sum of elements in given Matrix

ye ho jayega ?

**HW:** Find the maximum element in  ***Earnes*** given 2D array.

**Ques:** Find the row with maximum sum

|   | 0 | 1 | 2 | 3 |      |
|---|---|---|---|---|------|
| 0 | 9 | 4 | 8 | 4 | 25   |
| 1 | 6 | 1 | 1 | 5 | ✓ 59 |
| 2 | 2 | 2 | 3 | 6 | 13   |

**HW:** Find the minimum element out of all the maximum elements of each row

|   | 0 | 1 | 2 | 3 |   |
|---|---|---|---|---|---|
| 0 | 9 | 4 | 8 | 4 | 9 |
| 1 | 6 | 1 | 1 | 5 | 5 |
| 2 | 2 | 2 | 3 | 6 | 6 |

\* HW: Find the maximum element out of all the minimum of each col.

**Ques:** Print elements of 2D Array  
column-wise

# Ques: Print matrix in Snake Pattern Earn ₹s

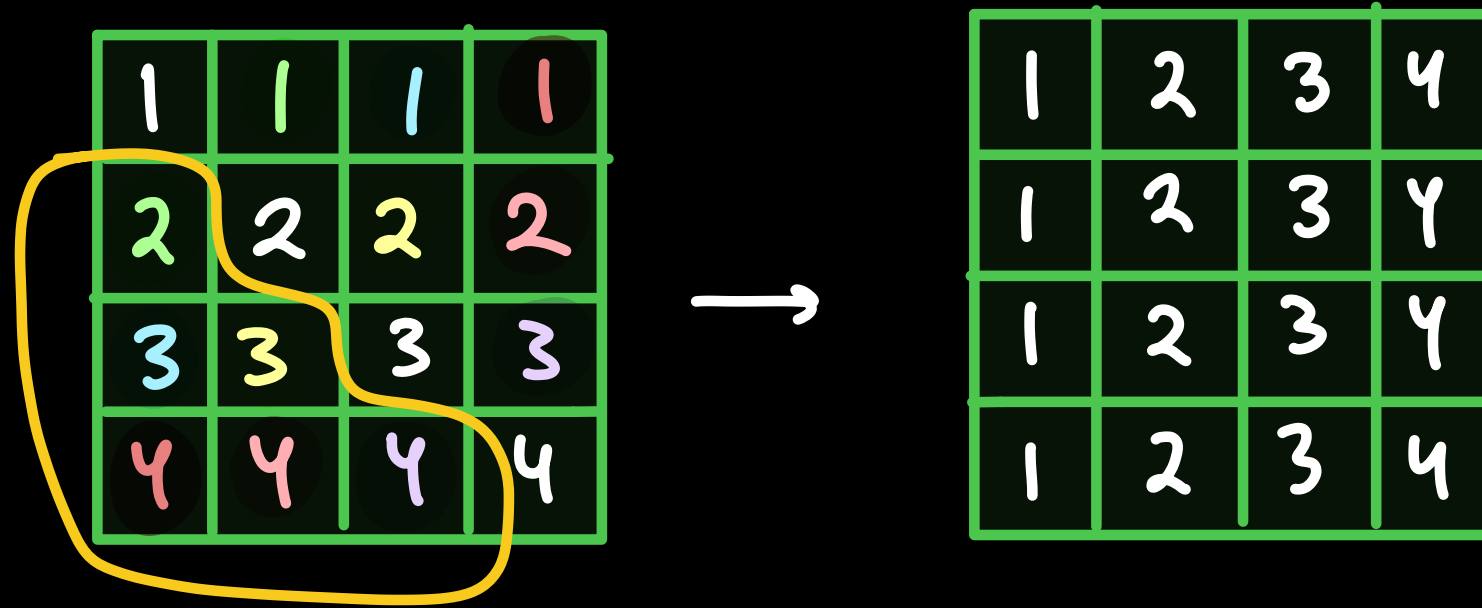
|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 6 | 0 | 9 | 2 | 3 |
| 1 | 5 | 2 | 6 | 0 | 9 |
| 2 | 8 | 4 | 8 | 1 | 7 |
| 3 | 4 | 1 | 7 | 3 | 5 |

if row is even, col  $\rightarrow$  0 to  $n-1$   
if row is odd, col  $\rightarrow$   $n-1$  to 0

Output. 6 0 9 2 3 9 0 6 2 5 8 4 8 1 7 5 3 7 1 4

# Ques: Transpose of Matrix

(hint: swap)  Earn ₹s



$\text{swap}(\text{arr}[i][j], \text{arr}[j][i])$

GFG